

By Prob 16(a), $A \cap B \subseteq B$. Since, By 2nd Commutative Law, $(A \cap B) = (B \cap A)$.
 By 1st Commutative Law, ~~$A \cap B$~~ $(A \cup B) = (B \cup A)$. By Prob 16(b),
 $B \subseteq A \cup B$. $\therefore |A \cap B| \leq |A|, |B| \leq |A \cup B|$.

$\therefore$ If, $|A \cap B| = |A \cup B|$, then $|A \cap B| = |A| = |B| = |A \cup B|$.

Claim: If $|A \cap B| = |A| = |B| = |A \cup B|$, then $A = B$

Proof: Suppose for the sake of contradiction, then $A \neq B$.
 By Defn. 2 from Sec-2.1, WLOG, let $x \in A$ but $x \notin B$. By Defn. of
 not belongs to, $\neg(x \in B)$. By Defn. of negation, $\neg(\neg(x \in B))$ is False. By
 Double-Negation Law of Propositions, $x \in B$ is False. By Defn. of OR, $x \in A \vee$
 $x \in B$ is ~~True~~ True. By Defn. 1, $x \in (A \cup B)$. $\therefore |B| < |A \cup B|$.

This is a contradiction. $\therefore A = B$ $\square$

d) $0 \leq J(A, B) \leq 1$ and $0 \leq d_J(A, B) \leq 1$

Solution: By Prob 16(a), $A \cap B \subseteq A$. By Prob 16(b), $A \subseteq A \cup B$.
 By Prob 19, from Ex-2.1, $A \cap B \subseteq A \cup B$. By Defn. 3 from Sec-2.1,
 if x is an arbitrary element of $A \cap B$, then $x \in (A \cup B)$.
 $\therefore |A \cap B| \leq |A \cup B| \Rightarrow \frac{|A \cap B|}{|A \cup B|} \leq 1$.

$|A \cap B| = 0$, when $A \cap B = \emptyset$ $\therefore 0 \leq \frac{|A \cap B|}{|A \cup B|} \leq 1 \Rightarrow 0 \leq J(A, B) \leq 1$

$1 - 0 \leq 1 - J(A, B) \leq 1 - 1 \Rightarrow 1 \geq d_J(A, B) \geq 0$ $\square$